

Review questions

- (01) Data - Raw facts, which can be processed to produce information
- Database - A collection of related data
- DBMS - A software system that enables users to define, create, maintain, and control access to the database
- Database system - A combination of a database, DBMS, and other software tools and applications to manage and use the database effectively
- Database Catalog - A repository of metadata that describes the structure of the database
- Program - Data - independence - The property that allows programs to be immune to changes in the data representation and storage
- User view - A subset of the database that is relevant to a particular user or application
- DBA - A person or group responsible for the administration and maintenance of the database system
- End user - A person who interacts directly with the database system to retrieve or manipulate data
- Canned transaction - A pre-programmed database transaction that performs a specific function or task
- Deductive Database system - A database system that incorporates logical inference and deduction capabilities

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Persistent object - An object whose existence is independent of the program that created it, typically stored in a database

Meta-data - Data that describes other data, such as data dictionary information or database schema

Transaction - processing application - An application that interacts with a database by performing transactions, which are sequences of operations that maintain the consistency

(02) Defining

This involves specifying the data types, structures, and constraints of the data to be stored in the database

Constructing

The process of storing the data on some storage medium that is controlled by the DBMS

Manipulating

This includes functions such as querying the database to retrieve specific data, updating the database to reflect changes in the miniworld

Sharing

A database allows multiple users and programs to access the database simultaneously

- (03) Self-describing nature of a database system
Insulation between programs and data, and data abstraction
Support of multiple views of the data
Sharing of data and multiuser transaction processing

01) Self-describing nature of a database system

In traditional file processing systems, the structure of the data is typically defined within the application programs themselves. This means that each application program is tightly coupled with a specific database, as the data structure is declared directly within the program's code.

02) Insulation between Programs and data

Database systems provide insulation between programs and data by implementing a layer of abstraction known as the data model. This abstraction shields application programs from the details of how data is stored and structured internally in the database.

In traditional file systems, there is often minimal insulation between programs and data. Application programs directly access and manipulate data stored in files using low level file operations. It also typically lack robust data abstraction mechanisms.

03) Support of multiple views of the data

Database systems support data abstraction, which allows users to interact with the database at different levels of abstraction. This includes providing multiple views of the data. Views can be created to provide each group with the relevant subset of data, hiding unnecessary details and improving data

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security. Views can also be created to provide each group with the relevant subset of data, hiding unnecessary details and improving data security.

(04) Sharing of data and multitransaction processing

Database systems allow multiple users and applications to share and access the same data concurrently. This capability enables collaboration and facilitates efficient data utilization across an organization. Multitransaction processing involves the execution of multiple transactions concurrently by different users and applications.

(04) DBA

Administering the resources

Responsible for authorizing access

Coordinating and monitoring its use

Acquiring software and hardware resources

Database designers

Identifying the data

Communicate with all prospective database users

Supporting the requirements of all user groups

(05) Casual End users - Occasionally access the database

Naive or parametric end users - Constantly querying and updating the database, using standard types of queries and updates

Sophisticated end users - familiarize themselves with the facilities of the DBMS in order to implement their own applications

standalone users - mostly maintain personal ~~addresses~~ databases
using ready-to-use packaged applications

06) Definition of a database

Data manipulation

Data querying and retrieval

Data security and access control

07)

Database

Information retrieval

- Structured data
- Efficient storage, retrieval and manipulation
- SQL

- Unstructured or semi-structured
- Finding information through user queries
- Keyword based searching or more more ranked algorithms

Exercise

01) Informal queries

Not all courses in the computer science department
Find all the students who received an A grade in
the full 2007 semester

What are the prerequisites for CS3380

Who is the instructor for MATH2410 offered in
spring 2008

Update operators

Change class of ~~somet~~ Smith to sophomore

Update credit hours for CS3320 to 3

Delete the record for MATH2410 course offered in fall 2007

02)

Controlled redundancy

- Planned and intentional duplication of data to improve performance

- Storing the department name (CS) in both the course and section tables

Uncontrolled

redundancy

- Unplanned and unintentional duplication of data which can lead to data integrity algorithms

- Storing student names and IDs in both ~~are~~ the student table and a separate spreadsheet maintained by an advisor

03) One to many
Many to many

04) For instructors - a view that shows all sections they are teaching along with the student names enrolled in those sections

For advisors - A view that shows all courses a student has taken and their grades in those courses

For department chairs - A view that shows the number of students enrolled in each course offered by their department

05) Primary key constraint

Student table $\rightarrow$ Student number

Foreign key constraint

Course no $\rightarrow$ Section table, Course table

06) Simple data structures - For small, flat files with a simple structure

Read only data - A simpler file system maybe sufficient

07) A) Course table

Department - This column would change from CS to CSSE

Course number - This column will be updated with new profile

B) Course table

Add a column named Course ID

Remove columns department and Course-no

Section table

Modify the Course-No $\rightarrow$ Course-ID (Course table)

Prerequisite table

Modify Course-No Prerequisite-no $\rightarrow$ Course-ID (Course table)